SRV Records: Specialised Services

In Part 5 — DNS A Records and Subdomains, we saw how A records connect a domain name to a server.
In Part 6 — Email DNS Records, we configured MX, SPF, DKIM, and DMARC to ensure that email is delivered correctly and authenticated properly.

For a website, you add an A record in DNS. For email, you add an MX record. These records map a domain name to a server.

SRV records address a different requirement.

Some specialised services, such as Microsoft Teams, VoIP phone systems, multiplayer game networks, XMPP (Jabber) messaging, and directory services such as Active Directory, require not just **where** to connect but also **how**: which **protocol** to use, which **port** to try, and whether backup servers exist. That's when SRV records are required.

An SRV record configured in DNS provides all of these values.

If this feels unfamiliar, that's normal. Most website owners will never configure an SRV record. You don't need one unless your provider specifically asks for it.

Core Terminology

SRV record — A DNS record that defines how a specific service is reached. It provides protocol, port, priority, weight, and target hostname.

Service label — The first part of the record name (for example, `_sip`). It identifies the service.

Protocol label — The second part of the record name (for example, `_tcp` or `_udp`). It defines the transport method.

Port — The numerical endpoint on a server where a service listens (for example, `5060` for SIP).

Priority — Determines failover order. Lower values are tried first.

Weight — Influences traffic distribution between records with equal priority.

Target — The hostname providing the service. It must resolve to an A or AAAA record and mustn't be a CNAME.

Beyond Web and Mail

Websites and email cover the majority of DNS use cases.

Specialised services, such as those listed in the introduction, rely on automatic discovery. That means:

- You publish connection information for the service in an SRV record in DNS.
- Client applications query DNS for that SRV record.
- They connect to the service automatically.

You don't need to manually configure each client application with server names, protocols, and ports. That would quickly become time-consuming, especially across a company.

The client application (for example, Teams) queries DNS in this format:

```
_service._protocol.yourdomain.com
```

For example:

```
_xmpp-client._tcp.yourdomain.com
_sip._tcp.yourdomain.com
_minecraft._tcp.yourdomain.com
```

The format uses two labels:

- The service being requested (for example, `_sip`)
- The transport protocol it expects (`_tcp` or `_udp`)

The leading underscores distinguish these service records from normal hostnames such as www or mail.

If the record exists, DNS returns the connection details that tell the application exactly how to connect. If it doesn't exist, automatic discovery fails, and the service may be unreachable.

An A record alone can't provide this information. A records only map a hostname to an IP address. They can't specify:

- Port numbers
- Transport protocol
- Priority values
- Traffic distribution rules

That's why SRV exists as a separate DNS record type.

You only publish an SRV record when your provider specifically requests you to.

Ports and Protocols

An IP address identifies a server. A port identifies a specific service running on that server.

Examples:

- 80 → HTTP
- 443 → HTTPS
- 25 → SMTP
- 5060 → SIP
- 5222 → XMPP

A single server can run many services at the same time. The IP address directs traffic to the correct machine. The port directs that traffic to the correct application process on that machine.

Transport protocols define how data is transmitted:

- TCP provides reliable, ordered delivery
- UDP prioritises speed and low latency

Many real-time communication services require a specific protocol and port to function correctly. If a client attempts to connect using the wrong protocol or port, the connection fails — even if the service itself is running.

For services that rely on automatic discovery, that level of detail is necessary. SRV records publish these additional connection parameters in DNS.

A typical SRV record looks like:

```
_service._protocol.yourdomain.com 3600 IN SRV priority weight port target
```

Example:

```
_sip._tcp.yourdomain.com 3600 IN SRV 10 50 5060 sip.provider.com
```

This record tells the client:

- Which service is it connecting to (`_sip`)
- Which protocol to use (`_tcp`)
- Which port the service listens on (`5060`)
- Which server to prefer (`sip.provider.com`)

Without SRV records, these values would need to be configured manually on every device.

Advanced Routing

Some services need multiple servers for backup or load balancing.

When a client queries `_service._protocol.yourdomain.com`, DNS can return multiple SRV records. Each response includes routing information that helps the client decide how to connect.

Priority

Priority controls failover behaviour. Lower numbers are tried first.

```
_sip._tcp.yourdomain.com 3600 IN SRV 10 50 5060 sip1.provider.com
_sip._tcp.yourdomain.com 3600 IN SRV 20 50 5060 sip2.provider.com
```

The client first attempts to connect to `sip1.provider.com`.
If that server isn't reachable, it then attempts `sip2.provider.com`.

This allows the service to keep running even if a primary server becomes unavailable.

Weight

When multiple records share the same priority, weight controls traffic distribution.

```
_sip._tcp.yourdomain.com 3600 IN SRV 10 60 5060 sip1.provider.com
_sip._tcp.yourdomain.com 3600 IN SRV 10 40 5060 sip2.provider.com
```

In this example, both servers have equal priority. The weight value determines how traffic is distributed.

Approximately:

- 60% of traffic goes to `sip1.provider.com`
- 40% of traffic goes to `sip2.provider.com`

This provides basic load distribution without additional infrastructure.

Target Requirements

The target:

- Must be a hostname

- Must resolve to an A or AAAA record
- Mustn't be a CNAME

Because the target is a hostname, you can change the underlying IP address later without modifying the SRV record itself. This separation simplifies maintenance and provider changes.

Now that you understand how SRV records work, let's walk through adding one to your domain.

Step 1 — Add the Record

In your DNS control panel:

- Record type: `SRV`
- Name/Host: `_sip._tcp`
- TTL: `3600`
- Priority: `10`
- Weight: `50`
- Port: `5060`
- Target: `sip.provider.com`

Make sure `sip.provider.com` resolves to a valid A or AAAA record.

It's sufficient to specify `_sip._tcp`. Your provider adds the `yourdomain.com` suffix automatically.

Step 2 — Verify the Record

Run:

```
nslookup -type=SRV _sip._tcp.yourdomain.com
```

Verify that:

- The SRV record resolves
- The target hostname resolves separately
- The port and priority values match your provider's instructions

Troubleshooting

Configuration mistakes are often small but disruptive.

If the service doesn't connect as expected, check the following:

- The target hostname resolves to an A or AAAA record
- The port value matches your provider's documentation
- `_service` or `_protocol` labels are typed correctly
- Priority values are configured as intended

In many cases, the application reports only a general connection error. The service itself may be running correctly, but the DNS configuration prevents clients from discovering it.

Conclusion

SRV records extend DNS beyond simple name-to-address mapping. They allow a domain to publish service-specific connection details, including protocol, port, priority, and server selection.

Most domains don't require them, and many hosting customers will never need to configure one. When a provider relies on automatic service discovery, however, SRV records become part of the connection process.

Correct configuration is essential for service discovery.

In the next tutorial, we examine Reverse DNS and its role in server reputation and deliverability.